

1.7 RESONANCE CONDITION

RF works only when its frequency matches the Larmor frequency

Resonance occurs when the frequency of the RF pulse (f_{RF}) exactly matches the Larmor frequency (f_0) of the protons. Only then can **maximum energy** be absorbed and the net magnetization be tipped effectively. If the frequencies do not match, little or no energy is absorbed.

THE CONCEPT

Protons are like children on a swing (magnetization M). Pushing the swing at the right rhythm ($f_{RF} = f_0$) makes it go higher (maximum excitation). Too slow or too fast... the swing barely moves.

Too slow
($f_{RF} < f_0$)
Low swing



Right rhythm
($f_{RF} = f_0$)
High swing



Too fast
($f_{RF} > f_0$)
Low swing

f_0 = Larmor frequency (Hz)
 f_{RF} = RF frequency (Hz)
 Δf = Frequency difference (Hz)
 $\Delta f = f_{RF} - f_0$

Resonance Condition:

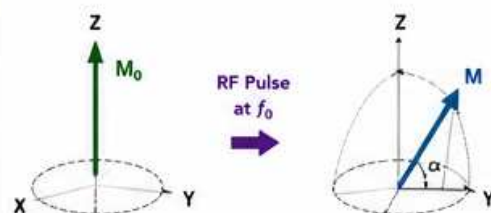
$$f_{RF} = f_0$$

$$\Delta f = 0$$

EFFECT OF FREQUENCY MATCHING

ON RESONANCE ($f_{RF} = f_0$)

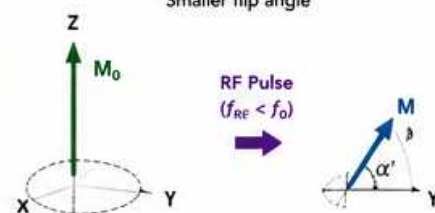
Maximum energy absorbed
Efficient tipping (desired flip angle)



Strong transverse magnetization → strong signal

OFF RESONANCE – TOO LOW ($f_{RF} < f_0$)

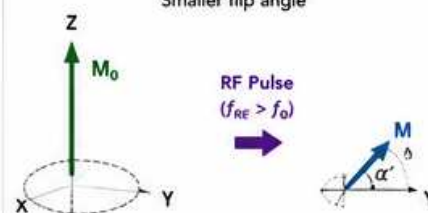
Protons cannot follow the slow RF
Less energy absorbed
Smaller flip angle



Weak transverse magnetization → weak signal

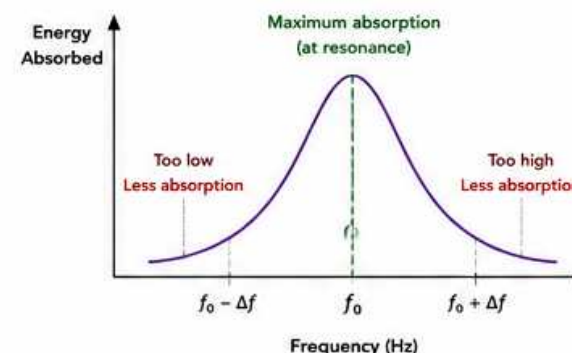
OFF RESONANCE – TOO HIGH ($f_{RF} > f_0$)

Protons cannot follow the fast RF
Less energy absorbed
Smaller flip angle



Weak transverse magnetization → weak signal

RESONANCE CURVE (ABSORPTION vs FREQUENCY)



Key Points

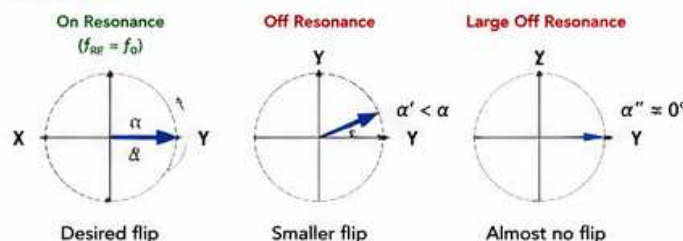
- Peak occurs at $f_{RF} = f_0$
- As $|\Delta f|$ increases, absorption drops.
- The width of the curve depends on B_1 strength (RF bandwidth).
- Keeping $f_{RF} = f_0$ across the patient is critical.

Stronger B_1 (wider RF bandwidth) → broader resonance curve
Weaker B_1 → narrow curve → more sensitive to frequency errors

WHAT HAPPENS WHEN OFF RESONANCE?

Phase and Magnitude Errors

- M is tipped less than intended.
- Transverse magnetization (M_{xy}) is reduced.
- Phase errors can occur, causing image artifacts.



RESONANCE IN THE MRI SYSTEM

- The scanner's RF transmitter is set to a specific frequency (f_{RF}).
- That frequency must match the Larmor frequency at the magnetic field strength (B_0).
- Local field variations (B_0 inhomogeneity) cause local f_0 variations → off resonance → signal loss or distortion.
- Shimming is used to make f_0 uniform across the imaging volume.

FIELD SERVICE TIP

- Verify the correct center frequency in service mode or protocol settings.
- Check B_0 homogeneity (shim values).
- Look for frequency drifts in RF system or temperature-related shifts.
- Use a phantom to confirm uniform excitation at the correct frequency.

ENGINEERING INSIGHT

At 1.5T, f_0 for Hydrogen $\approx$ 63.87 MHz. A frequency error of just 100 Hz can cause noticeable signal loss when B_1 is weak or bandwidth is narrow.



COMMON MISCONCEPTION

"A little off frequency is okay."
Reality: Even small Δf can reduce flip angle and cause artifact, especially for short RF pulses or high resolution imaging.

REAL-WORLD ANALOGY

Tuning a radio: If you are off the station frequency, the sound is weak or noisy. Only when you are exactly on the station do you get a clear signal.



KEY TAKEAWAY

Resonance happens only when $f_{RF} = f_0$. Matching the RF frequency to the Larmor frequency allows maximum energy absorption and effective magnetization tipping. This is the foundation of all MR signal generation.